19AI303 - Engineering Mechanics & Product Development

For AI & DS , AI & ML Students

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L (Lecture)              = 2
T (Tutorial)             = 0
P (Practical)            = 2
C (Credit)               = 3

CONTACT HOURS PER WEEK : 4
Total Contact Hours    : 60
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Course Objectives

Determine the resultant of forces.

Determine moment of inertia and radius of gyration of plane and solids.

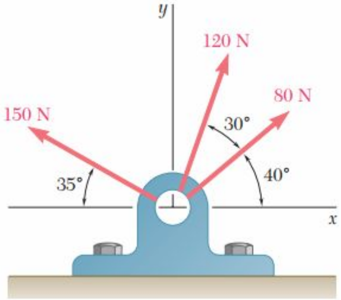
Demonstrate various concepts by working out problems relevant to real life applications of mechanisms.

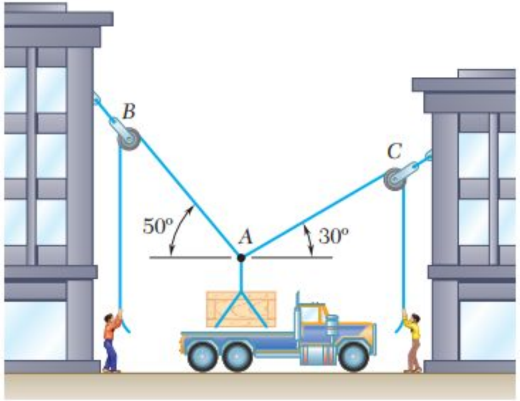
Understand the process to plan and develop product.

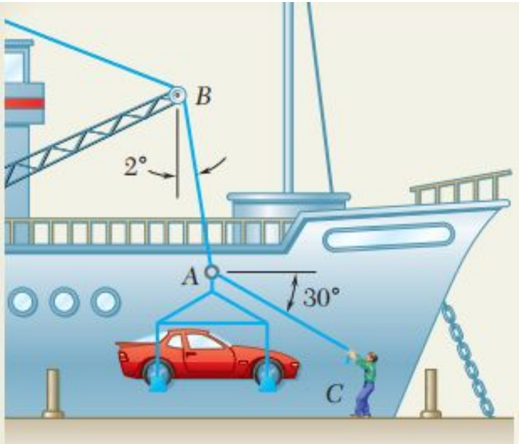
Create physical objects that satisfy product development/prototyping requirements using 3D Printing

Why 19AI303 ?

Mechanics - Tool to predict the future







Unit I
Forces and Particle Equilibrium

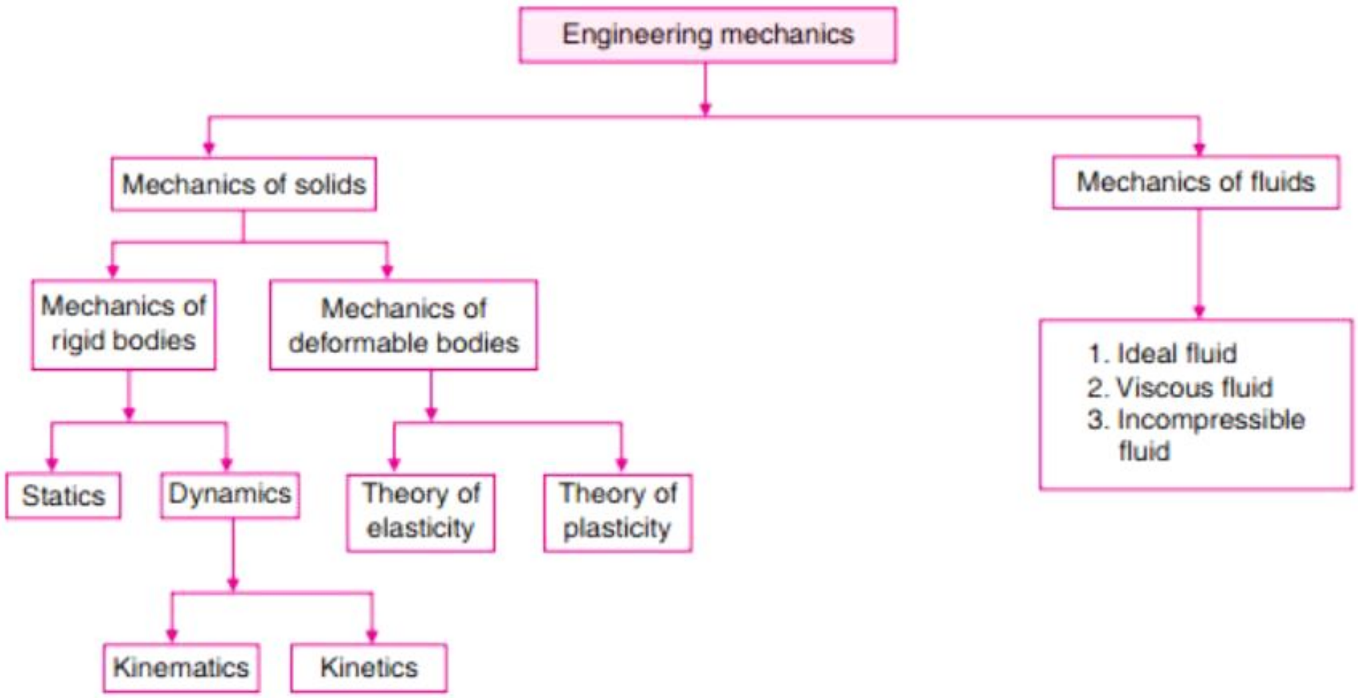
Laws of Mechanics – Lami's theorem, Parallelogram and triangular Law of forces - Vectorial representation of forces - Force Representation in 2D and Resultants - Force Representation in 3D - Particles Equilibrium.

Science

The branch of science, which coordinates the research work, for practical utility and services of the mankind, is known as Applied Science.

The subject of Engineering Mechanics is that branch of Applied Science, which deals with the laws and principles of Mechanics, alongwith their applications to engineering problems.

Branches



Engineering mechanics
Mechanics of solids
Mechanics of fluids
Mechanics of rigid bodies
Mechanics of deformable bodies
1. Ideal fluid
2. Viscous fluid
3. Incompressible fluid
Statics
Dynamics
Theory of elasticity
Theory of plasticity
Kinematics
Kinetics

Definitions

STATICS - which deals with the forces and their effects, while acting upon the bodies at rest.

DYNAMICS - which deals with the forces and their effects, while acting upon the bodies in motion. It is further subdivided into the following two branches :

1. Kinetics (which deals with the bodies in motion due to the application of forces)

2. Kinematics (which deals with the bodies in motion, without any reference to the forces which are responsible for the motion)

Units

There are only four systems of units, which are commonly used and universally recognised.

1. C.G.S. units
2. F.P.S. units
3. M.K.S. units
4. S.I. units

S.I. UNITS (INTERNATIONAL SYSTEM OF UNITS)

Prefixes

Factor by which the unit is multiplied	*Standard form*	*Prefix*	*Abbreviation*
1000 000 000 000	10^{12}	Tera	T
1 000 000 000	10^{9}	giga	G
1 000 000	10^{6}	mega	M
1 000	10^{3}	kilo	k
100	10^{2}	hecto*	h
10	10^{1}	deca*	da
0.1	10^{-1}	deci*	d
0.01	10^{-2}	centi*	c
0.001	10^{-3}	milli	m
0.000 001	10^{-6}	micro	µ
0.000 000 001	10^{-9}	nano	n
0.000 000 000 001	10^{-12}	pico	p

Standard abbreviations, which are internationally recognised

m for metre or metres

km for kilometre or kilometres

kg for kilogram or kilograms

t for tonne or tonnes

s for second or seconds

min for minute or minutes

N for newton or newtons

N-m for newton × metres (i.e., work done)

kN-m for kilonewton × metres

rad for radian or radians

rev for revolution or revolutions

Scalar Quantities

The scalar quantities (or sometimes known as scalars) are those quantities which have magnitude only such as length, mass, time, distance, volume, density, temperature, speed etc.

Vector Quantities

The vector quantities (or sometimes known as vectors) are those quantities which have both magnitude and direction such as force, displacement, velocity, acceleration, momentum etc.

The velocity of this cyclist is an example of a vector quantity.

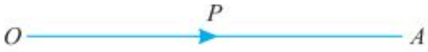


Fig. 1.2. Vector $\overrightarrow{OA}$

Important features of vector quantities

Representation of a vector

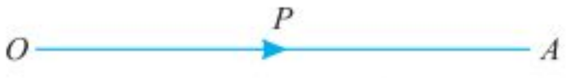


Fig. 1.2. Vector $\overrightarrow{OA}$

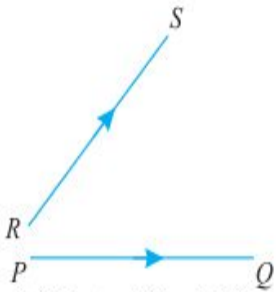
Unit vector -A vector, whose magnitude is unity, is known as unit vector.

Equal vectors - The vectors, which are parallel to each other and have same direction (i.e., same sense) and equal magnitude are known as equal vectors.

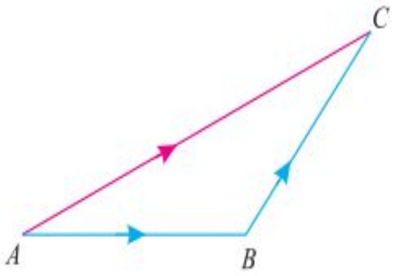
Like vectors - The vectors, which are parallel to each other and have same sense but unequal magnitude, are known as like vectors.

Important features of vector quantities

Addition of vectors

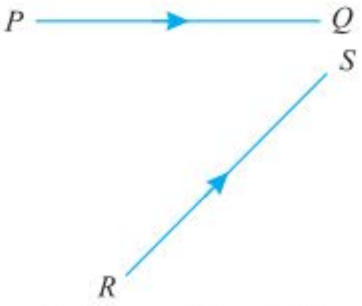


(*a*) Vectors *PQ* and *RS*

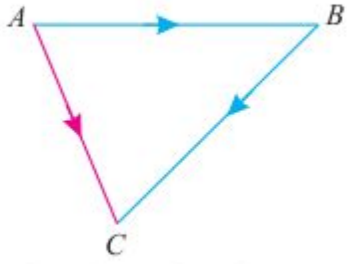


(*b*) Addition of vectors

Subtraction of vectors



(*a*) Vectors *PQ* and *RS*



(*b*) Subtraction of vectors

Laws of Mechanics

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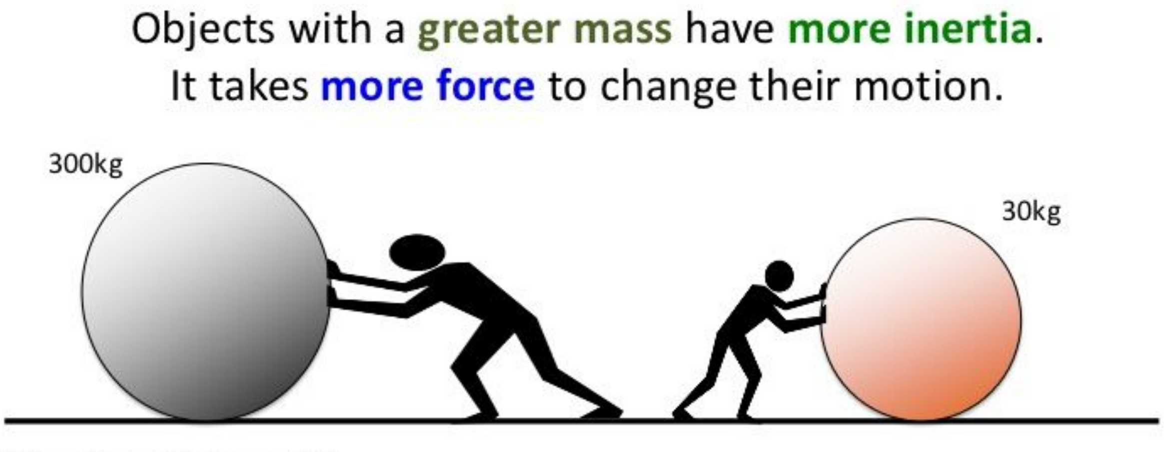
1. Newton’s first law
2. Newton’s second law
3. Newton’s third law
4. Newton’s law of gravitation
5. Law of transmissibility of forces, and
6. Parallelogram law of forces.

Newton's First Law

It states that every body continues in its state of rest or of uniform motion in a straight line unless it is compelled by an external agency acting on it.

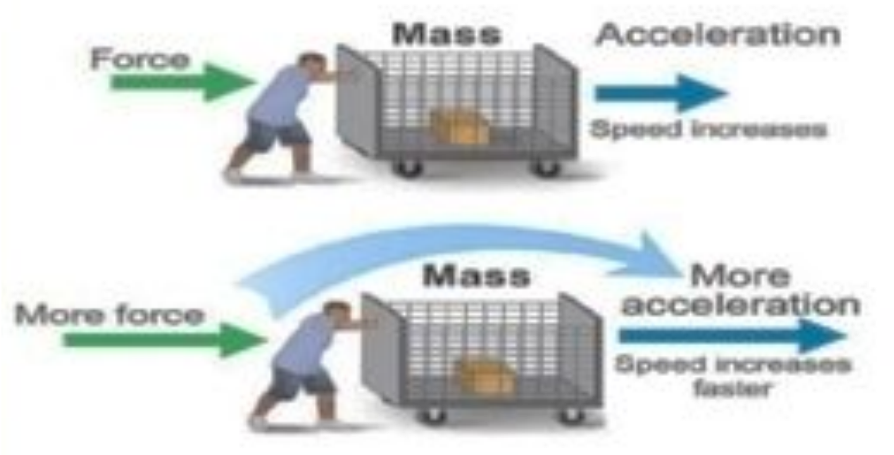
This leads to the definition of force as the external agency which changes or tends to change the state of rest or uniform linear motion of the body.



Newton’s Second Law

- It states that the rate of change of momentum of a body is directly proportional to the impressed force and it takes place in the direction of the force acting on it.
- Thus according to this law, Force = mass × acceleration

$F \propto m \times a$

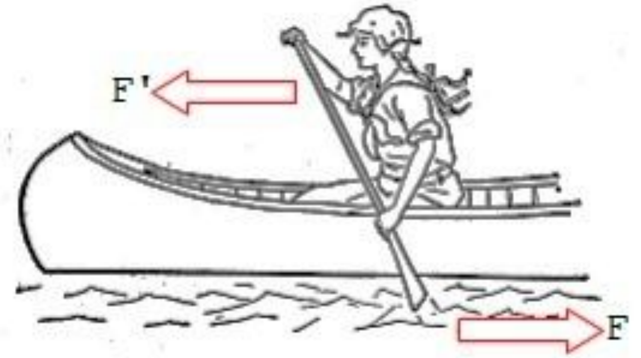


Newton’s Third Law

It states that for every action there is an equal and opposite reaction.

Consider the two bodies in contact with each other. Let one body applies a force F on another.

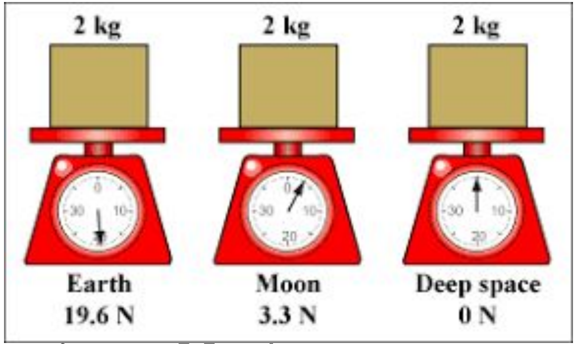
According to this law the second body develops a reactive force R which is equal in magnitude to force F and acts in the line same as F but in the opposite direction.



Important S.I Units

Sl.No.	*Derived Unit*	*Notation*
1	Area	m^2
2	Volume	m^3
3	Moment of Inertia	m^4
4	Force	N
5	Angular Acceleration	Rad/sec^2
6	Density	kg/m^3
7	Moment of Force	N.m
8	Linear moment	kg.m/sec
9	Power	Watt
10	Pressure/stress	$Pa(N/m^2)$
11	Mass moment of Inertia	$kg.m^2$
12	Linear Acceleration	m/s^2
13	Velocity	m/sec
14	Momentum	kg-m/sec
15	Work	N-m or Jule
16	Energy	Jule

Basic Terminologies in mechanics



Mass

- The quantity of the matter possessed by a body is called mass.
- When a body is taken out in a spacecraft, the mass will not change but its weight may change due to change in gravitational force.

Time

- Time is the measure of succession of events.
- The successive event selected is the rotation of earth about its own axis and this is called a day.

Displacement

- Displacement is defined as the distance moved by a body/particle in the specified direction.

Basic Terminologies in mechanics

Velocity

- The rate of change of displacement with respect to time is defined as velocity.

Acceleration

- Acceleration is the rate of change of velocity with respect to time.
- Thus $a = dv/dt$, where v is velocity

Momentum

- The product of mass and velocity is called momentum. Thus Momentum = Mass × Velocity

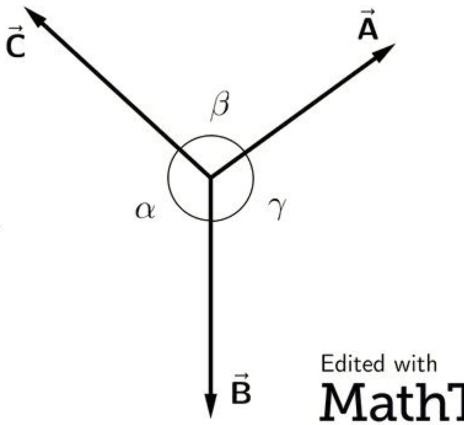
Rigid Body

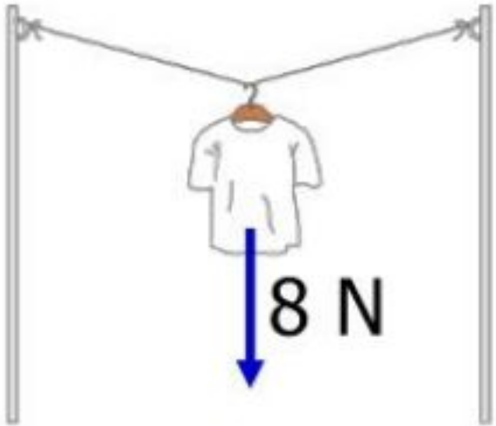
- A body is said to be rigid, if the relative positions of any two particles in it do not change under the action of the forces.

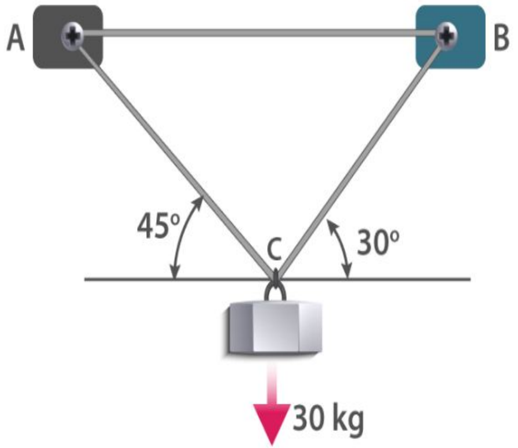
Lami's theorem

It states that, "If there forces acting at a point are in equilibrium, each force will be proportional to the sine of the angle between the other two forces."

$$\frac{A}{\sin\alpha} = \frac{B}{\sin\beta} = \frac{C}{\sin\gamma}$$

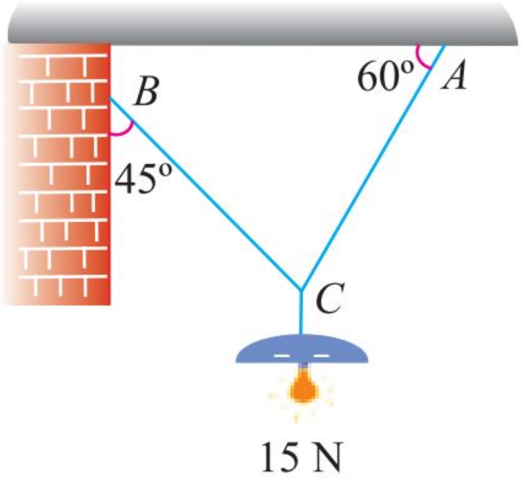






Problems in lamis theorem

An electric light fixture weighting 15N hangs from a point C, by two strings AC and BC. The string AC is inclined at 60° to the horizontal and BC at 45° to the horizontal as shown.



Solution. Given : Weight at $C = 15$ N

Let T_{AC} = Force in the string AC, and

T_{BC} = Force in the string BC.

The system of forces is shown in Fig. 5.4. From the geometry of the figure, we find that angle between T_{AC} and 15 N is 150° and angle between T_{BC} and 15 N is 135°.

$\therefore \quad \angle ACB = 180° - (45° + 60°) = 75°$

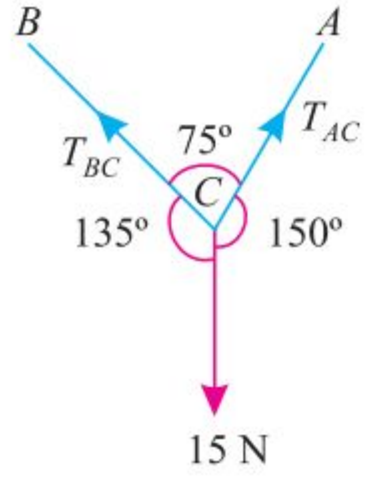


Fig. 5.4.

Applying Lami's equation at C,

$$\frac{15}{\sin 75°} = \frac{T_{AC}}{\sin 135°} = \frac{T_{BC}}{\sin 150°}$$

or

$$\frac{15}{\sin 75°} = \frac{T_{AC}}{\sin 45°} = \frac{T_{BC}}{\sin 30°}$$

$\therefore$

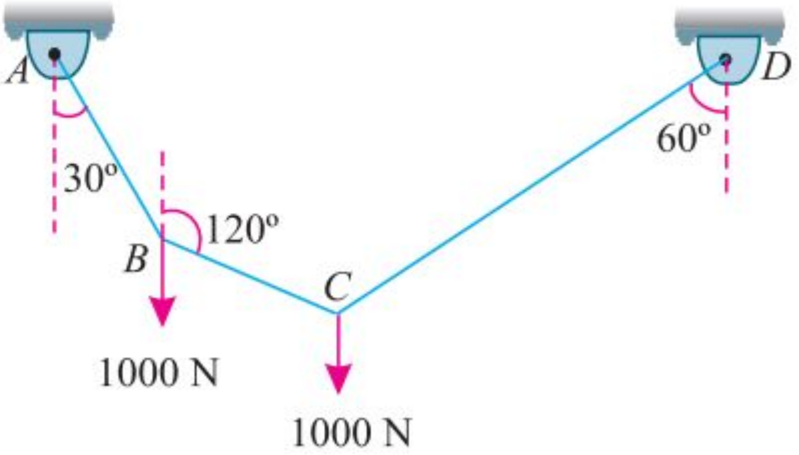
$$T_{AC} = \frac{15 \sin 45°}{\sin 75°} = \frac{15 \times 0.707}{0.9659} = 10.98 \text{ N} \quad \textbf{Ans.}$$

and

$$T_{BC} = \frac{15 \sin 30°}{\sin 75°} = \frac{15 \times 0.5}{0.9659} = 7.76 \text{ N} \quad \textbf{Ans.}$$

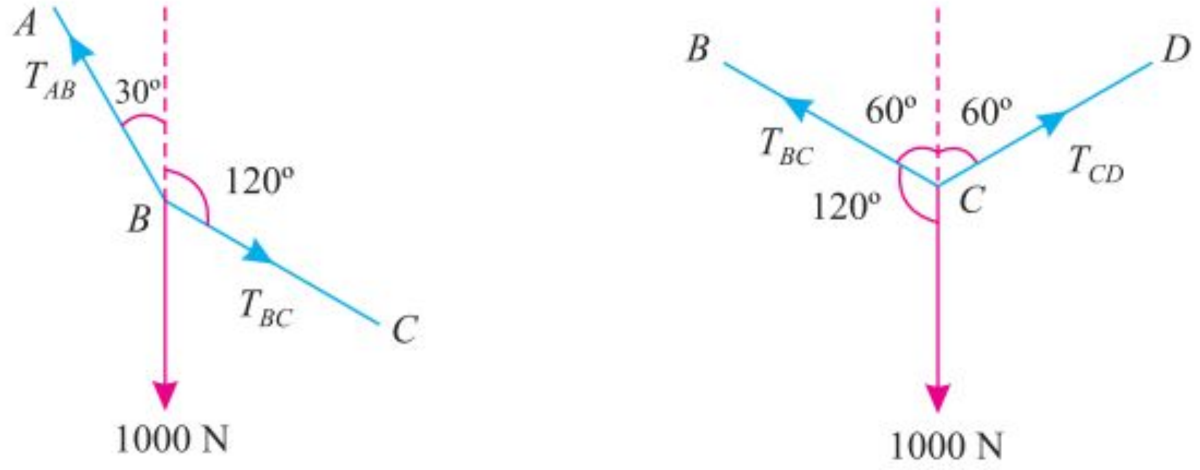
Problems in lamis theorem

A string ABCD, attached to fixed points A and D has two equal weights of 1000 N attached to it at B and C. The weights rest with the portions AB and CD inclined at angles as shown. Find the tensions in the portions AB, BC and CD of the string, if the inclination of the portion BC with the vertical is 120°



Solution. Given : Load at B = Load at C = 1000 N

For the sake of convenience, let us split up the string $ABCD$ into two parts. The system of forces at joints B and is shown in Fig. 5.6 (a) and (b).



(a) Joint B (b) Joint C

Fig. 5.6.

Let T_{AB} = Tension in the portion AB of the string,

T_{BC} = Tension in the portion BC of the string, and

T_{CD} = Tension in the portion CD of the string.

Applying Lami's equation at joint B,

$$\frac{T_{AB}}{\sin 60°} = \frac{T_{BC}}{\sin 150°} = \frac{1000}{\sin 150°}$$

$$\frac{T_{AB}}{\sin 60°} = \frac{T_{BC}}{\sin 30°} = \frac{1000}{\sin 30°} \qquad ...[\because \sin(180° - \theta) = \sin\theta]$$

$\therefore$
$$T_{AB} = \frac{1000 \sin 60°}{\sin 30°} = \frac{1000 \times 0.866}{0.5} = 1732 \text{ N} \quad \textbf{Ans.}$$

and
$$T_{BC} = \frac{1000 \sin 30°}{\sin 30°} = 1000 \text{ N} \quad \textbf{Ans.}$$

Again applying Lami's equation at joint C,

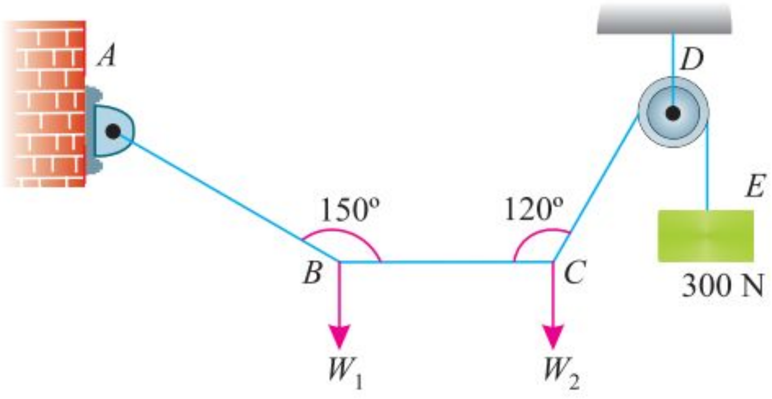
$$\frac{T_{BC}}{\sin 120°} = \frac{T_{CD}}{\sin 120°} = \frac{1000}{\sin 120°}$$

$\therefore$
$$T_{CD} = \frac{1000 \sin 120°}{\sin 120°} = 1000 \text{ N} \quad \textbf{Ans.}$$

Problems in lamis theorem

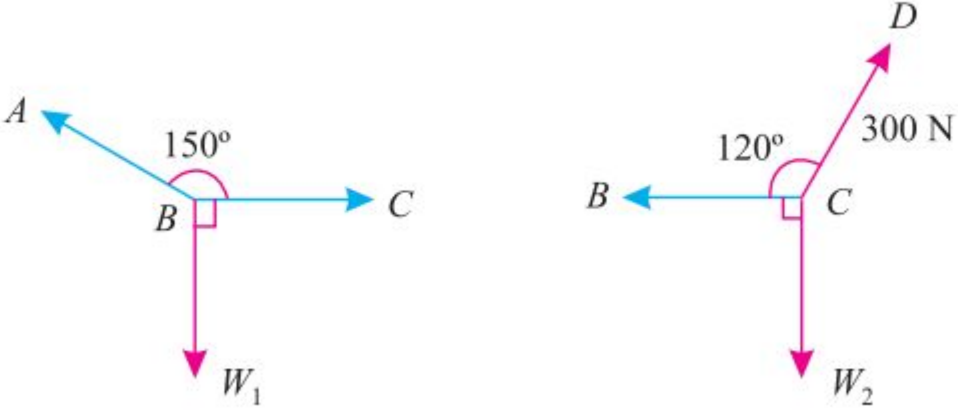
A light string ABCDE whose extremity A is fixed, has weights W1 and W2 attached to it at B and C. It passes round a small smooth peg at D carrying a weight of 300 N at the free end E as shown. If in the equilibrium position, BC is horizontal and AB and CD make 150° and 120° with BC,

Find (i) Tensions in the portion AB, BC and CD of the string and (ii) Magnitudes of W1 and W2



Solution. Given : Weight at $E = 300$ N

For the sake of convenience, let us split up the string $ABCD$ into two parts. The system of forces at joints B and C is shown in Fig. 5.8. (a) and (b).



(a) Joint B (b) Joint C

--- (*i*) *Tensions is the portion AB, BC and CD of the string*

Let T_{AB} = Tension in the portion AB, and

T_{BC} = Tension in the portion BC,

We know that tension in the portion CD of the string.

$$T_{CD} = T_{DE} = 300 \text{ N} \quad \textbf{Ans.}$$

Applying Lami's equation at C,

$$\frac{T_{BC}}{\sin 150°} = \frac{W_2}{\sin 120°} = \frac{300}{\sin 90°}$$

$$\frac{T_{BC}}{\sin 30^\circ} = \frac{W_2}{\sin 60^\circ} = \frac{300}{1} \qquad \text{...}[\because \sin(180^\circ - \theta) = \sin\theta]$$

$\therefore$ $$T_{BC} = 300 \sin 30^\circ = 300 \times 0.5 = 150 \text{ N} \quad \textbf{Ans.}$$

and $$W_2 = 300 \sin 60^\circ = 300 \times 0.866 = 259.8 \text{ N}$$

Again applying Lami's equation at B,

$$\frac{T_{AB}}{\sin 90^\circ} = \frac{W_1}{\sin 150^\circ} = \frac{T_{BC}}{\sin 120^\circ}$$

$$\frac{T_{AB}}{1} = \frac{W_1}{\sin 30^\circ} = \frac{150}{\sin 60^\circ} \qquad \text{...}[\because \sin(180^\circ - \theta) = \sin\theta]$$

$\therefore$ $$T_{AB} = \frac{150}{\sin 60^\circ} = \frac{150}{0.866} = 173.2 \text{ N} \quad \textbf{Ans.}$$

and $$W_1 = \frac{150 \sin 30^\circ}{\sin 60^\circ} = \frac{150 \times 0.5}{0.866} = 86.6 \text{ N}$$

EFFECTS OF A FORCE

It may change the motion of a body. i.e. if a body is at rest, the force may set it in motion. And if the body is already in motion, the force may accelerate it.

It may retard the motion of a body.

It may retard the forces, already acting on a body, thus bringing it to rest or in equilibrium.

It may give rise to the internal stresses in the body, on which it acts

PRINCIPLE OF PHYSICAL INDEPENDENCE OF FORCES

It states, “If a number of forces are simultaneously acting on a particle, then the resultant of these forces will have the same effect as produced by all the forces”

A particle may be defined as a body of infinitely small volume and is considered to be concentrated point.

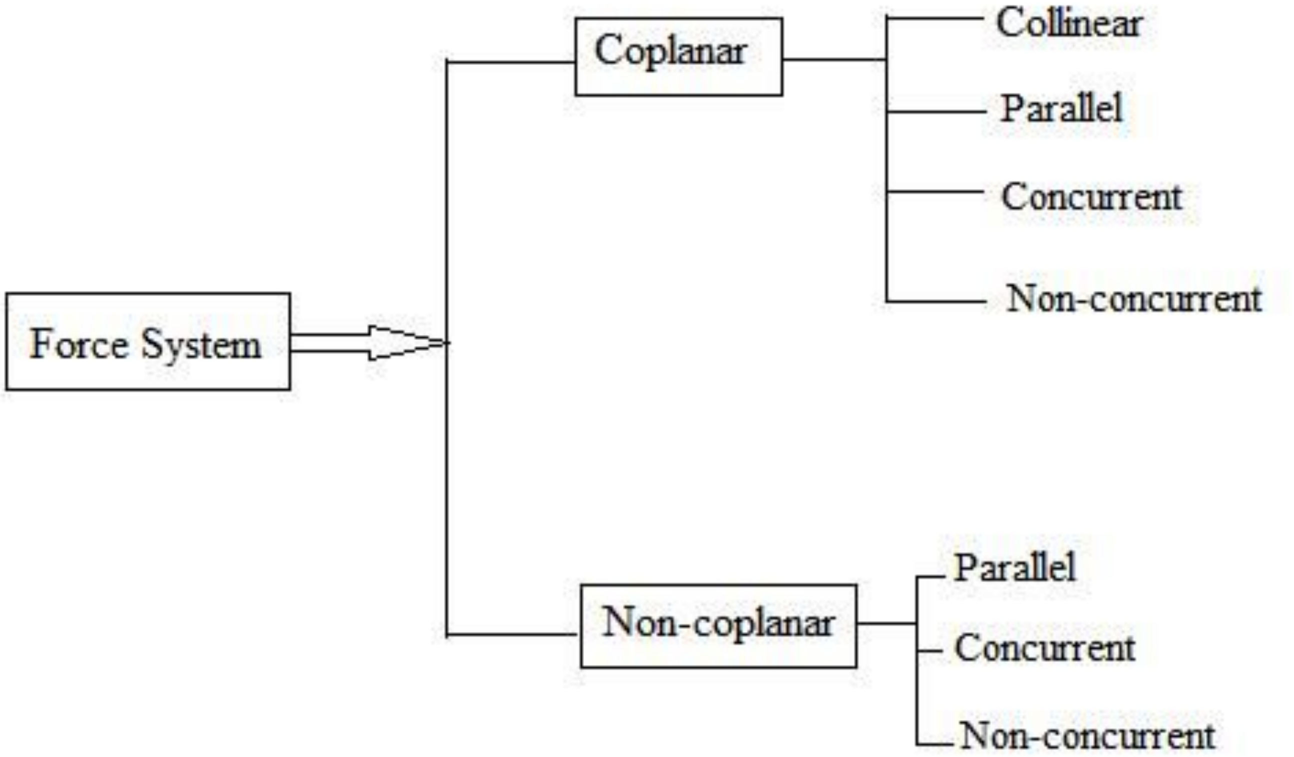
PRINCIPLE OF TRANSMISSIBILITY OF FORCES

It states, “If a force acts at any point on a rigid body, it may also be considered to act at any other point on its line of action, provided this point is rigidly connected with the body.”

A rigid body may be defined as a body which can retain its shape and size, even if subjected to some external forces. In actual practice, no body is perfectly rigid. But for the sake of simplicity, we take all the bodies as rigid bodies.

SYSTEM OF FORCES

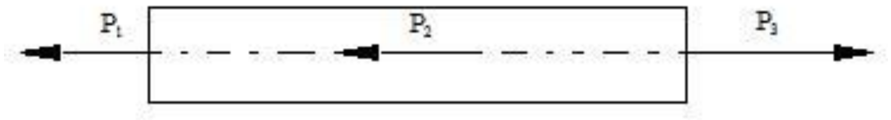
When two or more forces act on a body, they are called to form a system of forces



SYSTEM OF FORCES

Coplanar forces - The forces, whose lines of action lie on the same plane, are known as coplanar forces.

Collinear forces - The forces, whose lines of action lie on the same line, are known as collinear forces.



Concurrent forces - The forces, which meet at one point, are known as concurrent forces. The concurrent forces may or may not be collinear.The forces when extended pass through a single point and the point is called point of concurrency. The lines of actions of all forces meet at the point of concurrency.

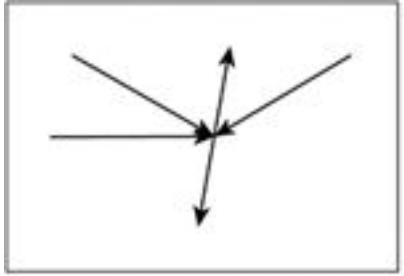
SYSTEM OF FORCES

Coplanar concurrent forces - The forces, which meet at one point and their lines of action also lie on the same plane, are known as coplanar concurrent forces.

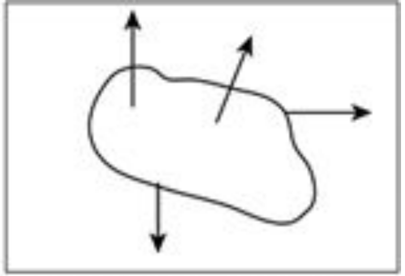
Coplanar non-concurrent forces - The forces, which do not meet at one point, but their lines of action lie on the same plane, are known as coplanar non-concurrent forces.

Non-coplanar concurrent forces - The forces, which meet at one point, but their lines of action do not lie on the same plane, are known as non-coplanar concurrent forces.

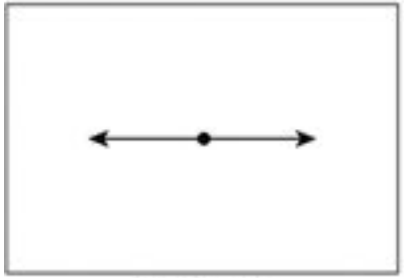
Non-coplanar non-concurrent forces - The forces, which do not meet at one point and their lines of action do not lie on the same plane, are called non-coplanar non-concurrent forces.



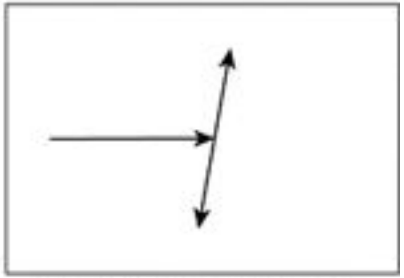
Coplanar concurrent



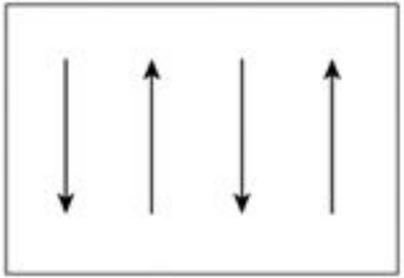
Coplanar non-concurrent



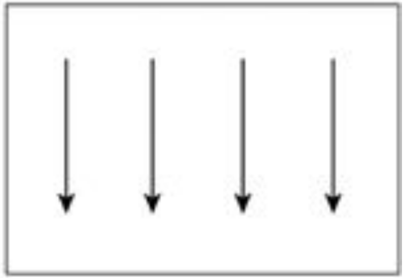
Collinear



Coplanar Non Collinear



Coplanar paraller



Coplanar like paraller